Machine Learning Report: Classification

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Preprocessing Overview:

Principal Component Analysis (PCA) was used to reduce the 3072 dimensions to 200 components that capture the most significant variance in the dataset, while reducing computational cost.

Report Task 3: Analysis for Classifiers

Single Decision Tree Classifier

For hyperparameter selection on the Decision Tree, I performed a Grid Search with 5-fold Cross-Validation on the training set to identify the optimal `max_depth` and `criterion`, by varying:

$$\text{max\_depth} \in \{5, 10, 15, 20\}, \qquad \text{criterion} \in \{\text{gini}, \text{entropy}\}.$$

The Best Decision Tree model identified via 5-fold cross-validation achieved:

- **Best Hyperparameters:** `max_depth = 10`, `criterion = gini`
- **CV score:** 0.304
- **Test accuracy:** 0.309
- **Number of Misclassified Images:** 6,914 out of 10,000

I also tested deeper Decision Trees (e.g., `max_depth` > 50), but the results tended to overfit, and so eventually I abandoned the configuration.

Figure 1: First 5 misclassified CIFAR–10 test images from the Decision Tree classifier.

A common misclassification pattern for the Single Decision Tree classifier is frogs being predicted as birds, suggesting that both their similar skin colour and the rough similarity of their shapes at low resolution mislead the classifier (e.g. image 3 the frog has a similar skin colour to a sparrow).

Ensemble Method (AdaBoost)

For the AdaBoost classifier, I used the same hyperparameters identified in the previous section and constructed an ensemble of 200 weak Decision Tree learners. I sampled 2,000 training points with replacement according to the current weights in each boosting iteration. The sample weights were updated after each iteration so that misclassified points received more weight for the next round. This ensures that the next weak learner focuses more heavily on correcting these misclassifications.

The AdaBoost Classifier identified via 5-fold cross-validation achieved:

- **Number of models: 200**
- **Sample size: 2000**
- **Hyperparameters:** `max_depth = 10`, `criterion = gini`
- **CV score:** 0.303
- **Test accuracy:** 0.372
- **Number of Misclassified Images:** 6,305 out of 10,000

I experimented with increasing the number of weak learners beyond 200 and the number of training samples, but the runtime rose sharply while the validation accuracy remained largely unchanged, so these configurations were abandoned.

Figure 2: First 5 misclassified CIFAR–10 test images from the AdaBoost classifier.

The common misclassification pattern for the AdaBoost Classifier should be similar to the Decision Tree Classifier since it inherits the weakness of the base model.

For instance, in image 4 the sample was previously misclassified as a bird by the Decision Tree classifier due to its skin colour, and AdaBoost similarly predicts it incorrectly as a deer.

This suggests that both models are influenced by the same colour-related features rather than the object shape.

Support Vector Machine (SVM)

For the SVM classifier, I performed a Grid Search with 3-fold cross-validation to tune the kernel type, the regularisation parameter C, and the RBF kernel parameter γ by varying:

$$C \in \{0.1, 1, 5\}, \qquad \text{kernel} \in \{\text{linear, RBF}\}, \qquad \gamma \in \{\text{scale}, 0.01, 0.001\}.$$

Note. From **scikit-learn**, setting $\gamma = \text{scale}$ automatically adjusts the RBF kernel parameter according to the input feature variance , where

$$\gamma_{\text{scale}} = \frac{1}{n_{\text{features}} \cdot \text{Var}(X)}$$

The best SVM model identified via 3-fold cross-validation achieved:

- **Best Hyperparameters:** $C = 5$, $\gamma = 0.01$, `kernel = RBF`
- **Best 3-fold CV Score:** 0.546
- **Test Accuracy:** 0.569
- **Number of Misclassified Images:** 4,306 out of 10,000

I discarded several SVM configurations, such as using 3-folds instead of 5-fold Cross-Validation, since it is computationally expensive. During the experiments, I observed that using the linear kernel resulted in a much higher runtime compared to the RBF kernel.

Figure 3: First 5 misclassified CIFAR–10 test images from the SVM classifier.

For the SVM misclassifications, it appears that the classifier relies heavily on background information. In image 3, for instance, the airplane is predicted as a ship, most likely due to the background resembling a maritime environment.

Report Task 4: Model Comparison and Discussion

Performance Comparison of the Three Classifiers

By comparing the test accuracy of each model, the SVM model performed the best by achieving 56.9% accuracy on the test set. While the Decision Tree classifier showed the weakest performance among the three models, it achieved only 30.9% test accuracy. On top of the Decision Tree, AdaBoost performed slightly better than the Decision Tree classifier because it combines many weak Decision Trees, reducing variance and allowing the ensemble to correct errors made by individual learners.

Model Complexity

Decision Tree: A single tree has limited representational power and relies on axis-aligned splits, making its decision boundaries too simple for complex image data.

AdaBoost: Boosting increases complexity by combining many weak learners, enabling more flexible boundaries, but each individual tree still has limited representational power for high-dimensional feature spaces.

SVM (RBF Kernel): The RBF kernel allows highly flexible, non-linear decision boundaries, giving the SVM outperforms Decision Tree and AdaBoost in complex high-dimensional feature spaces.

Bias–Variance Behaviour

Decision Tree: High variance and also high bias for non-axis-aligned patterns and easily overfits noise in high-dimensional data.

AdaBoost: Reducing variance through the aggregation of many weak learners compared to the Decision Tree Classifier, but the ensemble inherits the bias from the base learner.

SVM: Achieves a fairly strong balance between bias and variance due to the regularisation parameter C and the smoothness of the decision boundary enforced by the RBF kernel parameter γ.

Suitability for High-Dimensional Image Data

Decision Tree: Not suitable due to the loss of spatial structure after PCA, and often the classification of high-dimensional images depends on extensive features which is limited by the `max_depth` of the decision tree.

AdaBoost: The suitability is model-dependent since the performance is fundamentally affected by the base learner. In my experiment with using a Decision Tree as the base learner, it is better than a single tree, but it still inherits the lack of representation.

SVM: SVM is moderately suitable for High-Dimensional Image Data. It is flexible with respect to kernel choice, allowing the model to adapt its decision boundary depending on the structure of the data. It can capture the non-linear features for PCA-reduced, but the loss of spatial structure affects the performance compared to image-specific models such as CNNs.

Computational Characteristics

Decision Tree: Fastest to train but yields the lowest accuracy. It is computationally cheap but has weak performance.

AdaBoost: More expensive due to repeated training of weak learners. The runtime grows with ensemble size, but is manageable with a 2,000-sample bootstrap.

SVM: More computationally intensive than Decision Tree and AdaBoost, but PCA reduces cost. The RBF kernel offers strong accuracy within a reasonable training time than the Linear kernel.